

- What ARE the two forms of inverse variation?
- Based on yesterday: predict what $y = 1/x$ looks like.

Adult role

Solo teacher: circulate silently.

Do Now — Generalize: k in terms of x and y

13. Generalize: For an inverse variation, write an equation that gives the value of k in terms of x and y .

[CHECK] ANSWER KEY

Answer key (from TE): $k = xy$.

[SEARCH] PREDICTION (spend 30 sec)

Before we graph $y = 1/x$ together, sketch what you THINK it looks like. (No wrong answers — just predict.) We'll compare to the real thing in Launch.

LAUNCH — Reciprocal Function + Translations (teacher models Ex 4 + Ex 5)

Tag: teacher models Ex 4 + Ex 5 back-to-back

Launch

DOK: 2

Minutes: 12

Teacher does

- FIRST 6 min: Example 4 — graph $f(x) = 1/x$. Emphasize asymptote behavior (approaching but never reaching). Show table of $x \rightarrow 0$ and $x \rightarrow \pm\infty$.
- NEXT 5 min: Example 5 — graph $g(x) = 1/(x - 3) + 2$. Identify $(h, k) = (3, 2)$. Asymptotes shift accordingly.
- 30 sec: point at packet rule cards. "Mark them. These are your assessment reference."
- Do NOT solve Try-Its. Release at minute 17.

Students do

- Watch Ex 4. Compare your Do Now prediction to the real graph.
- Watch Ex 5. Note: h shifts horizontal, k shifts vertical. (h, k) is the new center.
- Copy BOTH rule cards into margins.

Questions to ask

- Why does $y = 1/x$ never equal 0? (Because $1/x = 0$ has no solution.)
- Why does $x \neq 0$ in the domain? (Division by zero.)
- For $g(x) = 1/(x - 3) + 2$: what's h ? What's k ? Where do asymptotes cross?
- What happens as $x \rightarrow 3$ from the right? From the left?

Adult role

Project both examples. Use a visible graphing tool (Desmos). Think aloud.

[MIC] TEACHER SCRIPT

1. “Yesterday we saw $y = k/x$. Today we graph it and name its features.”
2. Example 4: “As x gets huge (1000, 10000), y gets tiny (0.001, 0.0001) — approaching ZERO but never reaching. That’s a HORIZONTAL ASYMPTOTE at $y = 0$.”
3. “As x gets close to 0, y explodes — POSITIVE infinity from the right, NEGATIVE infinity from the left. That’s a VERTICAL ASYMPTOTE at $x = 0$.”
4. Example 5: “Now watch TRANSLATION. $g(x) = 1/(x - 3) + 2$. Subtract 3 inside → shift RIGHT
3. Add 2 outside → shift UP 2. New asymptotes: $x = 3$, $y = 2$.”
5. “The POINT (h, k) is the center of the translated graph. For $g(x)$, center is $(3, 2)$.”
6. Release.

[BOOK] RECIPROCAL FUNCTION — PARENT $f(x) = 1/x$

- Vertical asymptote: $x = 0$
- Horizontal asymptote: $y = 0$
- Domain: $\{x \mid x \neq 0\}$ • Range: $\{y \mid y \neq 0\}$
- Graph: hyperbola in Quadrants I and III (two branches, no intersection)

[BOOK] TRANSLATED RECIPROCAL — $g(x) = 1/(x - h) + k$

- Vertical asymptote shifts to: $x = h$
- Horizontal asymptote shifts to: $y = k$
- Domain: $\{x \mid x \neq h\}$ • Range: $\{y \mid y \neq k\}$
- The point (h, k) is the “new center” — where the asymptotes cross.
- Example: $g(x) = 1/(x - 3) + 2 \rightarrow$ asymptotes $x = 3$ and $y = 2$.

To find INTERCEPTS:

- y-intercept: set $x = 0$. $y = 1/(0 - h) + k = -1/h + k$.
- x-intercept: set $y = 0$. Solve $0 = 1/(x - h) + k \rightarrow x = h - 1/k$.

EXPLORE — Think-Pair-Share (35 min)

Tag: TPS — pure DOK-2 mastery — assessment rehearsal

Explore**DOK: 2****Minutes: 35****Teacher does**

- 6 laps across 35 min.
- Laps 1-3: Try-It 4 + #11 + #18. Parent-form scaled/reflected. (5-6 min each.)
- Laps 4-5: Try-It 5 + #19 — translated forms. #19 is the assessment rehearsal — press on BOTH asymptotes AND intercepts.
- Lap 6: #21 — modeling. If time.
- Do not give answers. Use sentence frame to press justification.

Students do

- 6 items in order. Parent forms first (Try-It 4, #11, #18), then translations (Try-It 5, #19, #21).
- For #19: BEFORE graphing, state (h, k) , asymptotes, and domain/range.
- Justify with sentence frame.

Questions to ask

- What's the center of the translation? (h, k)
- Which asymptote is which? ($x = h$ vertical; $y = k$ horizontal.)
- For #11: what y -value does $y = 5/x$ actually give at $x = 2$? (2.5 — the labeled 4 is wrong.)
- For #18: how does the negative flip the graph? (Quadrants II and IV instead of I and III.)
- For #19: what are the asymptotes? What are the intercepts?

Adult role

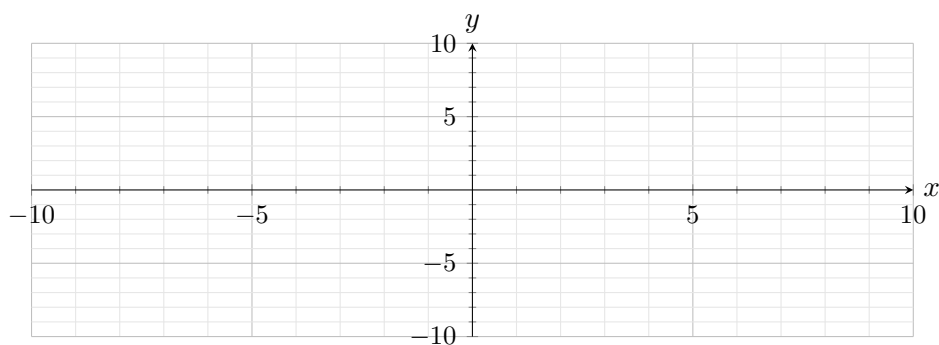
Solo teacher: 6 laps. Press BOTH asymptotes AND intercepts on #19. This IS the assessment rehearsal.

Bank-item labels (in order):

1. Try It 4 — Graph $g(x) = 10/x$ (scaled parent)
2. Practice #11 — Error Analysis: $y = 5/x$
3. Practice #18 — Graph $y = -2/x$ (reflected parent)
4. Try It 5 — Graph $g(x) = 1/(x + 2) - 4$ (translated)
5. Practice #19 [STAR] ASSESSMENT CRITICAL — $g(x) = 1/(x - 2) + 6$ + intercepts
6. Practice #21 — Video game storage (modeled)

Try It 4 — Graph $g(x) = 10/x$ (scaled parent)

Try It 4. Graph the function $g(x) = 10/x$. What are the domain, range, and asymptotes of the function?

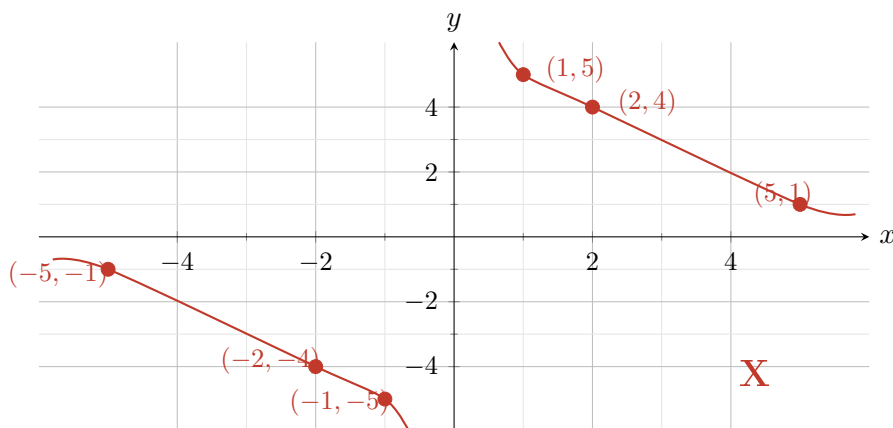


[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): graph passes through $(1, 10)$, $(10, 1)$, $(-10, -1)$, $(-1, -10)$; domain: all real numbers, $x \neq 0$; range: all real numbers, $y \neq 0$; horizontal asymptote $y = 0$; vertical asymptote $x = 0$. Graph of $g(x) = 10/x$ is a vertical stretch of $f(x) = 1/x$.

Practice #11 — Error Analysis: $y = 5/x$

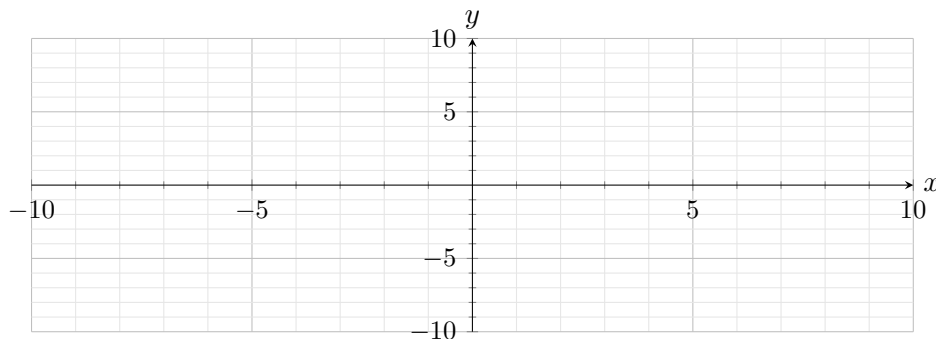
11. Error Analysis: Describe and correct the error a student made in graphing the function $y = 5/x$. The student's graph shows labeled points $(1, 5)$, $(2, 4)$, $(5, 1)$ in Quadrant I and $(-5, -1)$, $(-2, -4)$, $(-1, -5)$ in Quadrant III, with a large red X indicating the graph is incorrect.

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from TE): The points $(2, 4)$ and $(-2, -4)$ are not correct. The labels of these points should be $(2, 2.5)$ and $(-2, -2.5)$. Reason: $y = 5/2 = 2.5$, not 4.

Practice #18 — Graph $y = -2/x$ (reflected parent)

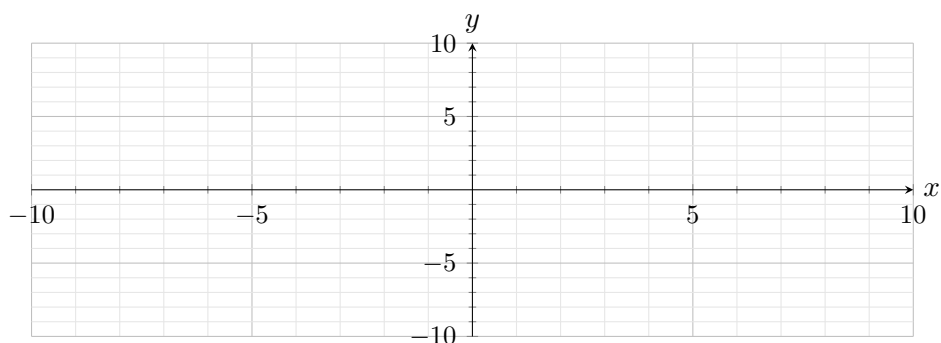
18. Graph the function $y = -2/x$. What are the domain, range, and asymptotes of the function?

**[CHECK] ANSWER / ANTICIPATED TALK**

Answer key (from TE): vertical asymptote: $x = 0$; horizontal asymptote: $y = 0$; domain: the set of real numbers except 0 ($x \neq 0$); range: the set of real numbers except 0 ($y \neq 0$). The graph is the reciprocal function reflected over the x -axis (Quadrants II and IV) and vertically stretched by a factor of 2.

Try It 5 — Graph $g(x) = 1/(x + 2) - 4$ (translated)

Try It 5. Graph $g(x) = 1/(x + 2) - 4$. What are the equations of the asymptotes? What are the domain and range?

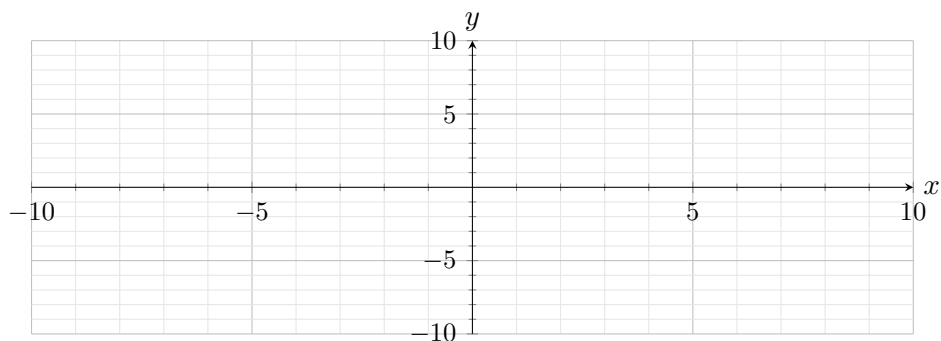


[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): vertical asymptote $x = -2$; horizontal asymptote $y = -4$; domain: set of real numbers with $x \neq -2$; range: set of real numbers with $y \neq -4$. Graph translated 2 units left and 4 units down from parent.

Practice #19 [STAR] ASSESSMENT CRITICAL — $g(x) = 1/(x - 2) + 6$ + intercepts

19. Graph $g(x) = 1/(x - 2) + 6$. What are the equations of the asymptotes? What are the domain and range? What are the intercepts?



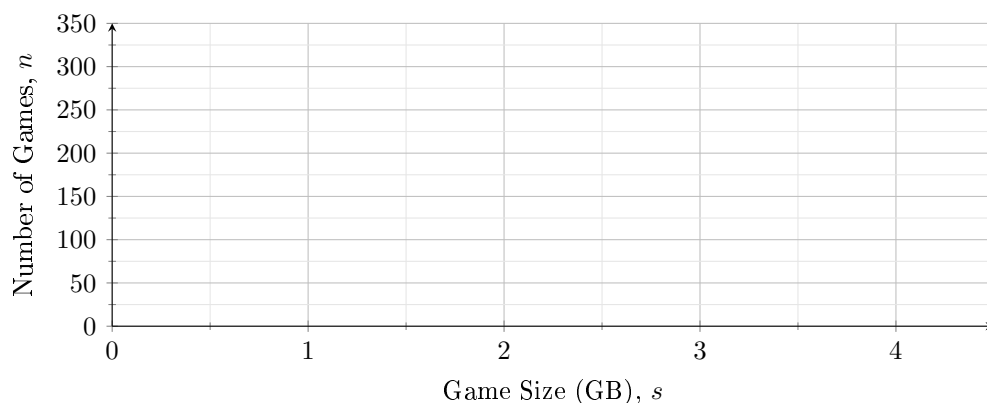
[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): vertical asymptote $x = 2$; horizontal asymptote $y = 6$; domain: set of real numbers with $x \neq 2$; range: set of real numbers with $y \neq 6$; y -intercept: 5.5 (or $11/2$); x -intercept: $11/6$. Graph shows translated reciprocal function with asymptotes intersecting at $(2, 6)$.

Practice #21 — Video game storage (modeled)

21. Use Structure: The number of downloaded games that can be stored on a video game system varies inversely with the average size of a video game. A certain video game system can store 160 games when the average size of a game is 2.0 gigabytes (GB). (a) Write an inverse equation that relates the number of games n that will fit on the video game system as a function of the average game size s in GB. (b) Use the inverse relationship to complete the table of values. Game Size (GB), s : 1.0, 2.5, 3.0, 4.0; Number of Games, n : $\underline{\quad}$, $\underline{\quad}$, $\underline{\quad}$, $\underline{\quad}$. (c) Sketch a graph of this inverse relationship on a coordinate plane.

Game Size (GB), s	1.0	2.5	3.0	4.0
Number of Games, n				



[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): (a) $n = 320/s$ (since $k = 160 \cdot 2.0 = 320$). (b) Table values: $s = 1.0 \rightarrow n = 320$; $s = 2.5 \rightarrow n = 128$; $s = 3.0 \rightarrow n = 106.7$; $s = 4.0 \rightarrow n = 80$. (c) Hyperbola through labeled points $(1, 320)$, $(2, 160)$ in Quadrant I and $(-1, -320)$, $(-2, -160)$ in Quadrant III (though physical context only supports Quadrant I).

SHARE / SUMMARY (5 min)

Tag: Concept Summary + LEHS 8-Q preview

Share / Summary

DOK: 2

Minutes: 5

Teacher does

- Cold-call 1 pair to present #19 (asymptotes + intercepts).
- Project Concept Summary (from student packet).
- “Topic 4 8-Q assessment: expect 2 of 8 questions on THIS content — asymptotes and translation descriptions. You just rehearsed both.”

Students do

- Presenting pair uses sentence frame for #19.
- Rest of class signals and annotates their Concept Summary.

Questions to ask

- What are the TWO big ideas from Lesson 4-1?
- If you see $g(x) = a/(x - h) + k$ on the assessment, what’s the FIRST thing you identify?

Adult role

Frame today as assessment rehearsal. Students leave confident.

Sentence frames

Pair share (#19): “This graph is the parent $f(x) = 1/x$ translated ___ units ___ and ___ units _____. The asymptotes are $x = ______$ and $y = ______$. The intercepts are _____.”

Whole class: one pair reads #19. Class signals thumbs up/sideways.

[CHART] CONCEPT SUMMARY (your reference for Topic 4 assessment)

INVERSE VARIATION ($y = k/x$):

- WORDS: As one variable increases, the other decreases; their product k is constant.
- ALGEBRA: $y = k/x$ where $k \neq 0$
- GRAPH: parent hyperbola, asymptotes $x = 0$ and $y = 0$

RECIPROCAL FUNCTION TRANSLATIONS ($y = a/(x - h) + k$):

- WORDS: The reciprocal function can shift left/right (h) and up/down (k).
- ALGEBRA: $y = a/(x - h) + k$
- GRAPH: asymptotes shift to $x = h$ and $y = k$; new “center” at (h, k) .

EXIT TICKET

Tag: pre-assessment confidence check

Exit Ticket

DOK: 1-2

Minutes: 3

Teacher does

- Collect at door.
- Tonight: scan confidence scores. If averages < 3 on asymptotes or translations, insert a 5-min Do Now recap when the 8-Q lands.

Students do

- Rate own confidence 1-5 on two skills.
- Name one thing to review.

Questions to ask

- Did any student score < 3 on either skill?
- What specific content did students name wanting to review?

Adult role

Collect. No grade. Use as data for future assessment prep.

[CLIPBOARD] EXIT TICKET — Student Form

I can identify asymptotes of $y = 1/(x - h) + k$. Confidence: 1 / 2 / 3 / 4 / 5

I can describe a translation from $f(x) = 1/x$ to $g(x)$. Confidence: 1 / 2 / 3 / 4 / 5

One thing I want to review before Topic 4 assessment: ____.

OPTIONAL REINFORCEMENT (not collected)

Extra stretch for fast finishers. Try without a calculator first.

Optional #1 (Savvas SAT/ACT style)

25. SAT/ACT. In a given set of experiments, t and w always vary inversely. When $t = 0.4$ seconds, $w = 172.8$ grams. How much time has passed when $w = 14.4$ grams?

- A. 2.4 s
- B. 4.8 s
- C. 27.648 s
- D. 995.328 s

[CHECK] ANSWER / ANTICIPATED TALK

Correct answer: B) 4.8 s (confirmed by TE answer key with filled circle on B).

[CLOCK] PERIOD A / PERIOD F — 65-MIN CLASS NOTES

Both sections should have 65 min. Use extra 10 min on Explore #19 (assessment-critical).
If finished early: Practice #25 (SAT/ACT MC) as fast-finisher stretch.
Alternative: project Practice #11 as a second Error Analysis prompt — more asymptote practice.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide both Rule cards as half-sheet cut-outs on their desks.
- For #19, provide a pre-axed grid (labeled x and y from -4 to 8) so they don't spend time drawing axes.
- Accept verbal asymptote identification in lieu of written.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: “asymptote” = line the graph approaches but never touches.
- “reciprocal” = the fraction flipped; $1/x$ is the reciprocal of x .
- “translation” = shift of the whole graph (no rotation or resize).
- Pair ELL students with English-dominant partners for TPS.